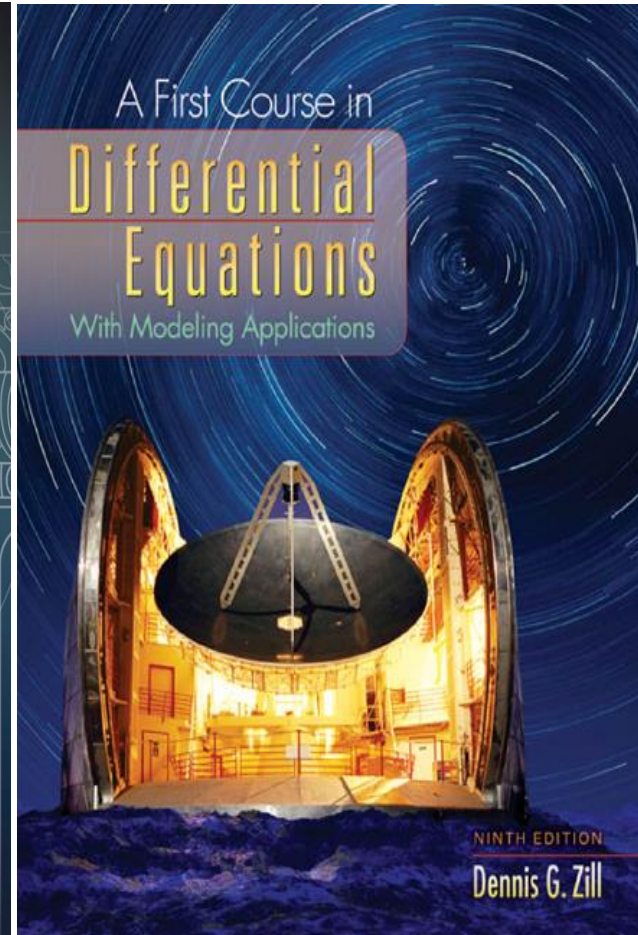
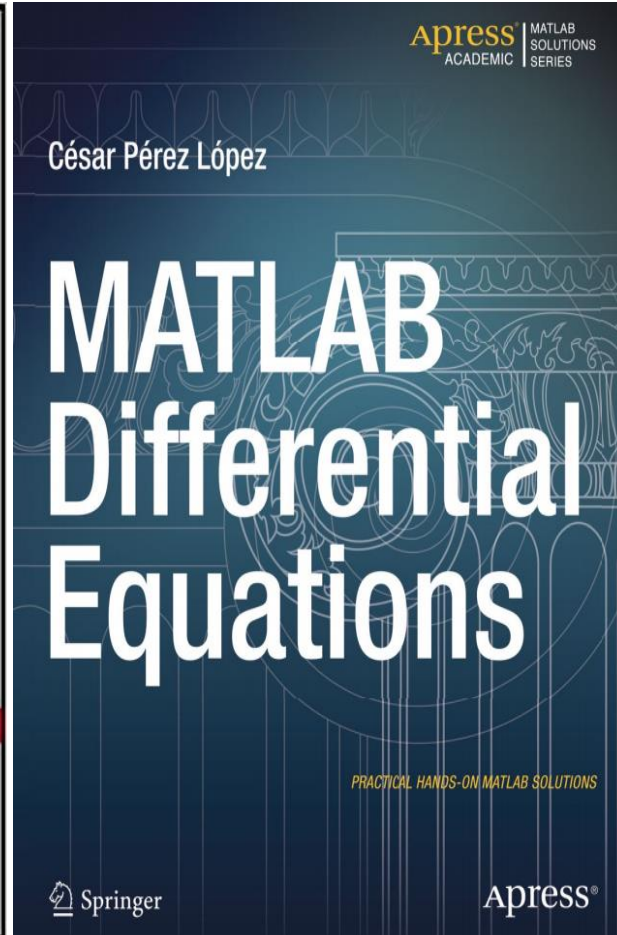
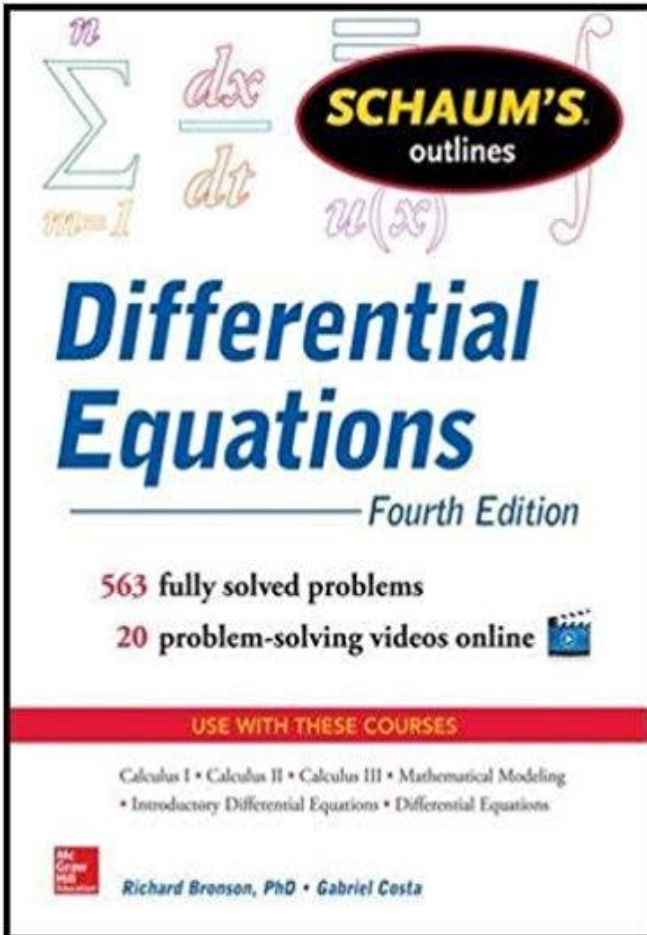


Differential Equations and Applications MTH203i



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Reference Books



Please see the course syllabus on the Moodle for more details

Course Description

This course is an introductory course in Ordinary Differential Equations (ODEs) and their engineering applications in different fields. The course focuses initially on finding the solutions of ODEs by different ways which cover analytical and numerical techniques for linear and nonlinear ODEs. Then the concept of time-frequency transformation has been introduced by Laplace transform followed by its ability to solve many ODEs easily. The modelling of different dynamical engineering applications has been discussed through the course as well as undergraduate research competition related to the importance of ODEs in science and engineering disciplines is applied by the end of the semester. Along the way students will study

- Basics of differential equations and their classifications.
- The solutions of first order differential equations.
- Modelling of dynamical engineering applications based on 1st ODEs.
- The solutions of higher order differential equations.
- Fourier series expansion of periodic functions
- Laplace transform technique: concept, properties, and solution of ODEs.
- The solution of linear systems of ODEs using Laplace transform.
- Solving PDEs using separation of variables method

Contents

- **Ch.1:- Introduction to Differential Equations**
- **Ch.2:- First-Order DE**
- **Ch.3:- Modeling with 1st order ODEs**
- **Ch.4:- Higher-Order DE**
- **Ch.5:- Fourier Series Expansion of Periodic functions**
- **Ch.6:- The Laplace Transform**
- **Ch.7:- Solving Partial differential Equations**

Course Grading

Midterm exam I	25%
Midterm exam II	
Tutorial quizzes	15%
Attendance (Lectures + Tutorials)	5%
Assignments (Tutorial)	15%
Final exam	40%

Attendance should be more than 75% (ByLaws)

Grading System (CS)

Letter grade	Grade point value	Percentage	Description
A+	4.0	=>97%	Excellent
A	4.0	=>90%	Excellent
A-	3.7	=>85%	Excellent
B+	3.3	=>80%	Very Good
B	3.0	=>75%	Very Good
B-	2.7	=>70%	Very Good
C+	2.3	=>65%	Good
C	2.0	=>62%	Good
C-	1.7	=>58%	Good
D+	1.3	=>55%	Pass
D	1.0	=>50%	pass
F	0.0	<50%	Fail

Rules

- ▶ Prof.'s and TA's are available only in the office hours which will be announced on the Moodle.
- ▶ There are JTA's will help you in the project.
- ▶ JTA will submit a periodic report on the progress of the team work in the project.
- ▶ HW should be submitted in a PDF form **only** (other formats are totally rejected).
- ▶ HW should be submitted on the submission link **only**.
- ▶ Each student should check his / her grades periodically per week.
- ▶ The complains for (HW – CW - ATT) grades are not allowed after the deadline announced of them (within **ONE** week).
- ▶ The complains for (Exams) grades will be in a certain time determined by Prof. / TA's with a deadline specified.

Revision on Differentiation and Integration

Sum and Difference Formulas

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Product to Sum Formulas

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

Double Angle Formulas

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

$$= 2 \cos^2 \theta - 1$$

$$= 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Half Angle Formulas

$$\sin^2 \theta = \frac{1}{2} (1 - \cos(2\theta))$$

$$\cos^2 \theta = \frac{1}{2} (1 + \cos(2\theta))$$

$$\tan^2 \theta = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$$

Revision on the Basic Functions

Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\operatorname{coth} x = \frac{\cosh x}{\sinh x}$$

$$\sinh(-x) = -\sinh x$$

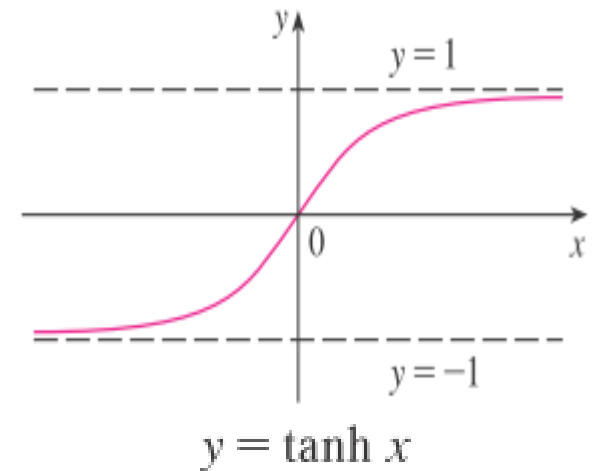
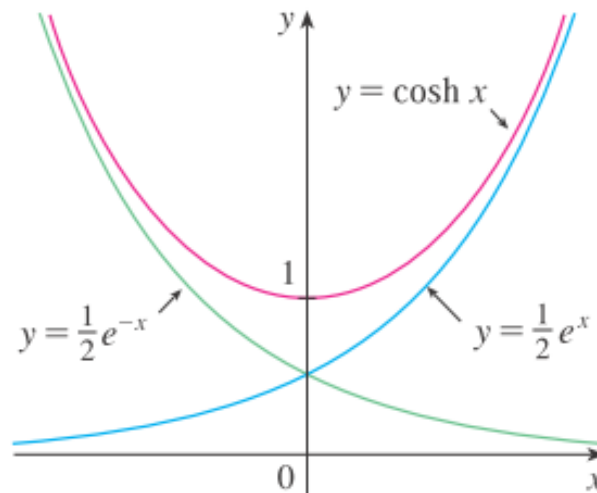
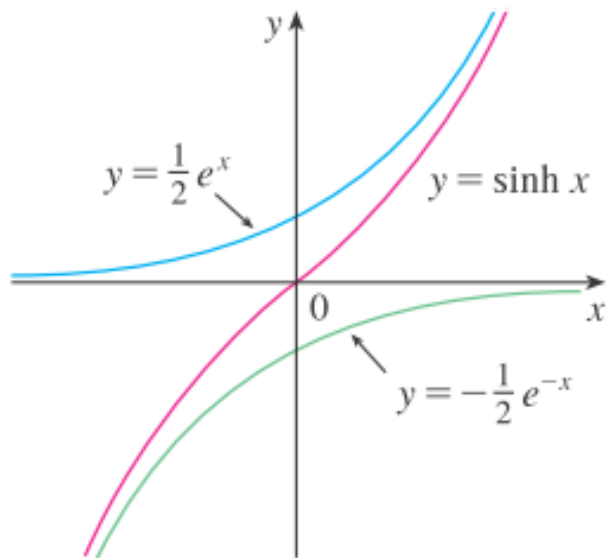
$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\cosh(-x) = \cosh x$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$



Revision on Integration

Basic Indefinite Integrals Rules

Rule (1)	$\frac{d}{dx} \int f(x) dx = f(x)$	$\int \frac{d f(x)}{dx} dx = f(x) + c$
Rule (2)	$\int k f(x) dx = k \int f(x) dx$	for any constant k
Rule (3)	$\int f(x) \pm g(x) dx = \int f(x) dx \pm \int g(x) dx$	
Rule (4)	$\int f(ax + b) dx = \frac{1}{a} F(ax + b) + c$ If $\int f(x) dx = F(x) + c$ and a, b are constants	
Rule (5)	$\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + c$	$, n \neq -1$
Rule (6)	$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$	
Rule (7) Substitution rule	$\int f(g(x)) g'(x) dx = \int f(u) du$ If $u = g(x)$ is differentiable function whose range is an interval I and f is continuous on I . Note that: $du = g'(x) dx$	

Revision on Integration

Basic Definite Integrals Rules

(1) Order Of Integration	$\int_a^b f(x) dx = - \int_b^a f(x) dx$
(2) Zero Width Interval	$\int_a^a f(x) dx = 0$
(3) Constant Multiple	$\int_a^b k f(x) dx = k \int_a^b f(x) dx$ for any number k $\int_a^b -f(x) dx = - \int_a^b f(x) dx$ $k = -1$
(4) Sum And Difference	$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$
(5) Additivity	$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$,where $a < b < c$
(6) Maximum And Minimum Inequality	If f has a maximum value $\max.f$ and $\min.f$ on $[a, b]$, then $\min.f.(b-a) \leq \int_a^b f(x) dx \leq \max.f.(b-a)$
(7) Domination	If $f(x) \geq g(x)$ on $[a, b] \Rightarrow \int_a^b f(x) dx \geq \int_a^b g(x) dx$ If $f(x) \geq 0$ on $[a, b] \Rightarrow \int_a^b f(x) dx \geq 0$ (Special case)
(8) Substitution Rule	$\int_a^b f(g(x)).g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$, $u = g(x)$ and $du = g'(x)dx$ If $g'(x)$ is continuous on the interval $[a, b]$ and f is continuous on the range of $g(x)$

Integration Techniques

- By Substitution
- Integration by Parts
- Trigonometric Substitution
- Trigonometric Integrals
- Partial Fractions

Differential Equations and Applications MATH 205 & MTH215



Chapter 1

Introduction to Differential Equations

Dr. Sameh Dakrouy

Chapter outlines

Introduction

First order ordinary differential equations:

Separable – Homogenous – Exact – Linear differential equations.

Equations reduce to separable or homogenous

Integrating Factors (equations reduce to exact)

Equations reduce to Linear

Definition

Differential equation is an equation that involves the derivative one variable (dependant variable) with respect to another variable (independent variable)

Examples

$$\frac{d^2y}{dx^2} + 5 \frac{dy}{dx} + 4y = 0$$

$$\frac{d^3y}{dx^3} + 2y \frac{dy}{dx} = y \ln x$$

**All derivatives are ordinary differentiation so it is called
Ordinary differential equation (ODE)**

When the DE involves partial derivatives it is called partial differential equation (PDE)

Examples

$$\frac{\partial U}{\partial x} - \frac{\partial U}{\partial y} = 4y - x$$

$$\frac{\partial^2 U}{\partial x^2} - \frac{\partial^2 U}{\partial y^2} = \sin(x + y)$$

**The equation involves partial differentiation so it is called
Partial differential equation (PDE)**

Order of Differential Equation

The order of the DE equals to the highest derivative in the equation.

- **Examples**

$$\frac{d^2y}{dx^2} + 4y = 0 \quad 2^{nd} \text{ order}$$

$$\frac{\partial^3 U}{\partial x^3} - \frac{\partial U}{\partial y} = (x^2 + y^2) \quad 3^{rd} \text{ order}$$

$$\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} = y \quad 2^{nd} \text{ order}$$

Solution of Differential Equation

- **Examples**

Show that $y = c_1 e^{2x} + c_2 e^{-2x}$ is the solution of the 2nd order ODE: $y'' - 4y = 0$

Answer: $y' = 2c_1 e^{2x} - 2c_2 e^{-2x}$

$$y'' = 4c_1 e^{2x} + 4c_2 e^{-2x}$$

Substitute in the D.E.: L.H.S.: $y'' - 4y = 4c_1 e^{2x} + 4c_2 e^{-2x} - 4(c_1 e^{2x} + c_2 e^{-2x}) = 0 = R.H.S.$

$\therefore y = c_1 e^{2x} + c_2 e^{-2x}$ is the solution of the O.D.E. $y'' - 4y = 0$

Finding the D.E. given the general solution

- **Example: Find the D.E. whose general solution is**

$$y = c_1x + c_2e^x$$

Answer:

$$y = c_1x + c_2e^x \dots(1) \quad , \quad y' = c_1 + c_2e^x \dots (2) \quad , \quad y'' = c_2e^x \dots (3)$$

From (1) , (2) and (3) eliminate c_1 and c_2

$$\text{From (3) in (2) : } y' = c_1 + y'' \rightarrow c_1 = y' - y''$$

$$\therefore y = (y' - y'')(x) + y''$$

the Differential equation is : $(1 - x)y'' + xy' - y = 0$

First Order Ordinary Differential Equations

The first order ODEs may appear in one of the following forms:

$$\begin{aligned}\frac{dy}{dx} &= f(x, y) \\ F(x, y, \frac{dy}{dx}) &= 0 \\ M(x, y)dx + N(x, y)dy &= 0\end{aligned}$$

The Solution of first order ODE is to find a relation in the form:

$$G(x, y, C) = 0$$

There is no one general analytical method for solving the first order ODEs. Therefore, the first order ODEs can be classified according to the method of solution to the following types:

First order First degree ODEs

- Separable D.Eqns - reduce to separable D.Eqns
- Homogeneous D. Eqns - Equations reduce to homogeneous
- Exact D.Eqns - Equations reduce to Exact (Integrating factors)
- Linear D. Eqns - Equations reduce to linear

Separable Differential Equations

The first order ODE is separable if it can be written in the form :

$$\frac{dy}{dx} = f(x)/g(y)$$

Or : $M_1(x)M_2(y)dx + N_1(x)N_2(y)dy = 0$

Solving the equation is by separating variables :

$$f(x) dx = g(y)dy$$

Integrating both sides

$$\int f(x) dx = \int g(y)dy$$

$$F(x) = G(y) + C$$

Example 1

Solve the D.E. $\frac{dy}{dx} = \frac{\sec^2 y}{1+x^2}$

Solution:

The equation can be written as:

$$\frac{dy}{\sec^2 y} = \frac{dx}{1+x^2}$$

$$\cos^2 y \, dy = \frac{dx}{1+x^2}, \quad \text{integrating both sides:}$$

$$\int \cos^2 y \, dy = \int \frac{dx}{1+x^2}$$

$$\int \frac{1}{2} (1 + \cos 2y) \, dy = \int \frac{dx}{1+x^2}$$

$$\frac{y}{2} + \frac{\sin 2y}{4} = \tan^{-1} x + C$$

Example 2

Solve the D.E. $(xy + 2y + 2x + 4)dx + x(x + 1)dy = 0$

Solution: by factorizing: $(y(x + 2) + 2(x + 2))dx = -x(x + 1)dy$

Take common factor $(x + 2)$:

$$(x + 2)(y + 2)dx = -x(x + 1)dy$$

Separating variables:

$$\frac{(x + 2)}{x(x + 1)} dx = -\frac{1}{(y + 2)} dy$$

Integrating: $\int \frac{(x+2)}{x(x+1)} dx = -\int \frac{1}{(y+2)} dy$ (apply partial fractions)

$$\int \left(\frac{2}{x} + \frac{-1}{x+1} \right) dx = -\int \frac{1}{(y+2)} dy$$

$$2 \ln x - \ln(x + 1) = -\ln(y + 2) + C \text{ (can be written as } \ln C \text{)}$$

$$\frac{x^2}{x + 1} = \frac{C}{y + 2}$$

Example3:- $y' = \sqrt{\frac{y^2 - 4y + 5}{11 - 10x - x^2}}$

Solution:

$$\int \frac{dy}{\sqrt{y^2 - 4y + 5}} = \int \frac{dx}{\sqrt{11 - 10x - x^2}}$$

$$\int \frac{dy}{\sqrt{(y - 2)^2 + 1}} = \int \frac{dx}{\sqrt{36 - (x + 5)^2}}$$

$$\sinh^{-1}(y - 2) + c = \sin^{-1}\left(\frac{x + 5}{6}\right)$$

Initial Value Problem (IVP)

When the differential equation was subjected to initial condition the whole problem is called IVP:

$$\frac{dy}{dx} = f(x, y) \text{ subject to } y(x_0) = y_0$$

Example: Solve the IVP $\frac{dy}{dx} = 8x^3 e^{-2y}, y(0) = 2$

Solution: The D.E. is separable: $e^{2y} dy = 8x^3 dx$

Integrating both sides: $\int e^{2y} dy = \int 8x^3 dx$

$$\frac{1}{2} e^{2y} = 2x^4 + C \text{ at } x = 0, y = 2$$

$$C = 0.5 e^4$$

$$\frac{1}{2} e^{2y} = 2x^4 + 0.5 e^4$$

Equations reduce to separable

The D.E. y' (or x') $= f(ax + by + c)$ can be reduced to separable by substitution $u = ax + by + c \rightarrow du = adx + bdy$

Example: Solve the D.E. $y' = \tan^2(x + y + 2)$

Solution : Let $u = x + y + 2 \rightarrow du = dx + dy \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$

$$\frac{du}{dx} - 1 = \tan^2 u$$

$$\frac{du}{dx} = 1 + \tan^2 u = \sec^2 u \text{ separable D.E.}$$

$$\cos^2 u \, du = dx \quad \text{integrating both sides}$$

$$\int \frac{1}{2} + \frac{1}{2} \cos 2u \, du = x + C$$

$$\frac{u}{2} + \frac{1}{4} \sin 2u = x + C \rightarrow \frac{(x + y + 2)}{2} + \frac{1}{4} \sin 2(x + y + 2) = x + C$$

Example:- Solve the following DEs

$$y' = (x + y + 1)^2$$

$$\text{Let } z = (x + y + 1) \Rightarrow z' = 1 + y' \Rightarrow y' = z' - 1$$

$$z' - 1 = (z)^2$$

$$z' = 1 + z^2 = \frac{dz}{dx}$$

$$\frac{dz}{1 + z^2} = dx$$

$$\tan^{-1} z = x + c$$

$$z = \tan(x + c)$$

$$\text{let } z = x + y + 1$$

$$y = \tan(x + c) - x - 1$$

Example:- Solve the following DEs

$$(x - y)dx + (y - x - 2)dy = 0, \quad y(0) = 2$$

$$y' = \frac{x - y}{x - y + 2}$$

$$\text{Let } z = (x - y) \rightarrow z' = 1 - y' \rightarrow y' = 1 - z'$$

Substitute in the DE

$$y' = \frac{x - y}{x - y + 2} \rightarrow 1 - z' = \frac{z}{z + 2}$$

$$1 - z' = \frac{z}{z + 2}$$

$$\frac{dz}{dx} = 1 - \frac{z}{z + 2} = \frac{2}{z + 2}$$

$$\int (z + 2)dz = \int 2dx \rightarrow z^2 + 4z = 4x + c$$

$$(x - y)^2 + 4(x - y) = 4x + c, \quad y(0) = 2 \rightarrow c = -4$$

Example:- Solve the following DEs

$$3y' + 2 = e^{-(2x+3y)} / (2x + 3y)$$

Let $z = (2x + 3y) \Rightarrow z' = 2 + 3y'$

$$z' = e^{-z} / z$$

$$\frac{dz}{dx} = \frac{1}{ze^z}$$

$$\int ze^z dz = \int dx$$

$$ze^z - e^z = x + c$$

$$e^{(2x+3y)}(2x + 3y - 1) = x + c$$

End of Lect. 1